

BHARAT PETROLEUM MANAGEMENT SYSTEM

Product Requirements Document (PRD)

Version 1.0 | May 2025 | Confidential

Based on BRD v1.2

Document Type	Product Requirements Document (PRD)
Project	Bharat Petroleum Management System (BPMS)
Version	1.0
Status	Draft
Prepared By	Product Team
Reviewed By	TBD
Approved By	TBD
Date Created	May 2025
Based On	BRD v1.2 — May 2025

This document is intended for internal use only.

1. Introduction

1.1 Purpose

This Product Requirements Document (PRD) translates the approved Business Requirements Document (BRD v1.2) for the Bharat Petroleum Management System (BPMS) into detailed, actionable product specifications. It defines what the system must do, how each user role interacts with it, which screens are accessible per role, and the functional behaviour expected from every module.

1.2 Product Vision

BPMS is an enterprise-grade, web-based management platform that eliminates paper-based operations at Bharat Petroleum outlets. The system centralises fuel stock tracking, nozzle readings, worker management, shift scheduling, expense logging, sales analytics, and reporting under a single role-aware interface accessible on desktop and tablet browsers.

1.3 Scope

Phase 1 of BPMS covers:

- Role-based access for Super Admin, Admin, and Worker
- 10 functional modules with dedicated page-level access control
- Real-time dashboard with role-tailored widgets
- Multi-lingual support (English + Hindi in Phase 1)
- Secure JWT-based authentication and RBAC enforcement

1.4 Definitions & Acronyms

Term	Full Form	Meaning
BPMS	Bharat Petroleum Management System	The software product described in this document
BRD	Business Requirements Document	Source document for this PRD
PRD	Product Requirements Document	This document
RBAC	Role-Based Access Control	Permission system based on user roles
INR	Indian National Rupee	Currency used for all financial values
i18n	Internationalization	Framework for multi-language UI support
JWT	JSON Web Token	Stateless authentication mechanism

DXA	Device-independent eXchange unit	Unit used for document formatting (technical)
-----	----------------------------------	---

2. User Roles & Access Control

BPMS enforces a three-tier Role-Based Access Control (RBAC) model. Every route, API endpoint, and UI component is gated by role validation. The table below summarises each role's access mode and the pages they can access.

2.1 Super Admin

SUPER ADMIN — Complete Access (Read-Only Mode)		
Access Mode: Read-Only — Can view all data across all modules. Cannot create, edit, or delete any record. Has exclusive privilege to create/delete Admin and Worker accounts.		

#	Page / Module	Access Description
1	Dashboard	View all real-time widgets: fuel gauges, today's sales, active workers, low-stock alerts, expense totals, fuel difference indicators, nozzle status.
2	Fuel Management	View current and historical stock levels for Petrol and Diesel. View refill logs, stock difference reports, and low-stock alerts.
3	Nozzle Management	View all nozzle readings (opening/closing) per shift and nozzle. View auto-calculated fuel sold differences.
4	Worker Management	View all worker profiles, assignments, shift allocations, and status. Can create and delete Admin and Worker accounts.
5	Daily Entries	View all daily entry records across all workers, shifts, and dates. Filter by date, worker, fuel type, or nozzle.
6	Shift Management	View shift definitions, assigned workers, total fuel sales, and cash collections per shift.
7	Expense Tracking	View all recorded expenses by category, date, and amount. View expense totals by period.
8	Generator Management	View generator fuel allocation records, consumption sessions, usage duration, and difference calculations.
9	Sales Analysis	View aggregated sales data: quantity sold, revenue, rate history, shift-level and daily-level breakdowns.
10	Reports	View and export daily, monthly, and worker-level reports. Cannot modify report content.

2.2 Admin

ADMIN — Complete Access (Edit Mode)

Access Mode: Full Edit — Can view, create, edit, and delete all operational records. Cannot delete Super Admin accounts. Manages day-to-day operations.

#	Page / Module	Access Description
1	Dashboard	View all real-time widgets. Can acknowledge and dismiss alerts. Sees operational summary for current day and shift.
2	Fuel Management	Create, edit, and delete stock entries. Log refill deliveries. Configure low-stock alert thresholds. View and resolve difference flags.
3	Nozzle Management	Record opening and closing readings per shift per nozzle. Edit readings. View auto-calculated fuel-sold figures.
4	Worker Management	Create, edit, activate/deactivate Worker accounts. Assign fuel types, shifts, and profiles. Cannot delete Super Admin accounts.
5	Daily Entries	Create, edit, and delete daily entry records for any worker or shift. Filter by date, worker, fuel type, or nozzle.
6	Shift Management	Define, edit, and delete shifts. Assign workers to shifts. View shift-level sales and cash collection summaries.
7	Expense Tracking	Create, edit, and delete expense records. Categorise expenses. View expense totals and history.
8	Generator Management	Log generator fuel allocation and consumption sessions. Edit records. View usage and difference reports.
9	Sales Analysis	View all sales analytics. Update fuel rates (Petrol/Diesel per litre). View rate history and shift/daily revenue breakdowns.
10	Reports	Generate, view, and export daily, monthly, and worker reports. Can filter before exporting.

2.3 Worker

WORKER — Limited Self-Service Access

Access Mode: Self-Service — Can view their own records and submit entries for their assigned tasks only. Cannot view other workers' records or any admin-level data.

#	Page / Module	Access Description
1	Dashboard	View a personal dashboard showing current stock level (assigned fuel type), their shift status, and personal daily sales summary.
2	Nozzle Management	View their assigned nozzle(s). Submit opening and closing readings for their shift. Cannot edit other workers' nozzle records.
3	Check In	Record shift check-in and check-out with timestamp. View their own check-in history for current day.
4	Shift Management	View their assigned shift details: start time, end time, fuel type, and nozzle. Cannot modify shift definitions.
5	My Attendance	View their own attendance history. See past check-in/check-out records, shift summaries, and absence logs.
6	Sales Entry	Submit fuel sales entries for their assigned nozzle and shift. Input quantity sold. Cannot view other workers' sales records.

2.4 Role–Page Access Matrix

The matrix below provides a consolidated view of all pages and which roles can access them, and in what mode.

Page / Module	Super Admin	Admin	Worker
Dashboard	✓ Read-Only	✓ Full Edit	✓ Personal View
Fuel Management	✓ Read-Only	✓ Full Edit	✗ No Access
Nozzle Management	✓ Read-Only	✓ Full Edit	✓ Own Nozzle Only
Worker Management	✓ Read-Only + Account Mgmt	✓ Full Edit	✗ No Access
Daily Entries	✓ Read-Only	✓ Full Edit	✗ No Access
Shift Management	✓ Read-Only	✓ Full Edit	✓ View Own Shift
Expense Tracking	✓ Read-Only	✓ Full Edit	✗ No Access
Generator Management	✓ Read-Only	✓ Full Edit	✗ No Access
Sales Analysis	✓ Read-Only	✓ Full Edit	✗ No Access
Reports	✓ Read-Only	✓ Full Edit	✗ No Access
Check In	✗ No Access	✗ No Access	✓ Own Only
My Attendance	✗ No Access	✗ No Access	✓ Own Only

Sales Entry	X No Access	X No Access	✓ Own Nozzle/Shift
-------------	-------------	-------------	--------------------

3. Functional Requirements — Module Specifications

3.1 Dashboard

The Dashboard is the landing page after login. Its content is role-filtered.

Widget / Element	Super Admin & Admin	Worker
Petrol Stock Gauge	Live litres — system-wide	Live litres — assigned fuel type only
Diesel Stock Gauge	Live litres — system-wide	Not shown (if assigned to Petrol)
Today's Sales	Total revenue + litres for all shifts	Personal shift revenue + litres
Active Workers	Count of workers currently checked in	Not shown
Low Stock Alert	Banner when stock < configured threshold	Not shown
Expenses Today	Running total of all expenses recorded	Not shown
Fuel Difference	Petrol & Diesel variance indicators	Not shown
Nozzle Status	Reading status of all 4 nozzles	Status of assigned nozzle only
My Shift Info	Not applicable	Current shift name, start time, assigned nozzle

3.2 Fuel Management

Accessible by Super Admin (read) and Admin (edit). Workers have no access.

Feature	Specification
Stock Display	Real-time current stock for Petrol and Diesel in litres
Opening Stock Entry	Admin records opening stock at start of each day
Refill / Delivery Logging	Admin logs refill quantity, supplier, timestamp; stock auto-increases
Automatic Deduction	Stock auto-decreases on each validated sales entry

Low Stock Alert Config	Admin sets threshold per fuel type; alert triggers on dashboard when breached
Difference Calculation	System computes Expected Stock – Actual Stock; non-zero flags for review
Stock History Log	Full timestamped audit trail of every stock change with user attribution
Separate Petrol / Diesel	Petrol and Diesel are tracked as independent tanks with no cross-use

3.3 Nozzle Management

Super Admin views all records. Admin can fully edit. Workers can submit readings for their assigned nozzle only.

Feature	Specification
Nozzle Configuration	4 nozzles: A (Petrol), B (Petrol), C (Diesel), D (Diesel)
Opening Reading	Recorded at shift start by assigned worker or Admin
Closing Reading	Recorded at shift end by assigned worker or Admin
Fuel Sold Calculation	Auto-calculated: Closing Reading – Opening Reading
Worker Assignment	Each nozzle is assigned one worker per shift
Shift Linkage	Every reading is linked to a specific shift and date
Edit Restriction (Worker)	Workers can only submit/edit readings for their own assigned nozzle
Difference Flag	System flags nozzle-level discrepancies against tank stock deductions

3.4 Worker Management

Super Admin can view all records and create/delete accounts. Admin can create and edit Worker accounts only. Workers have no access to this module.

Feature	Specification
Worker Profile Fields	Full Name, Mobile, Address, Photo, Worker ID (auto), Fuel Type, Shift, Joining Date, Status
Total Workers	System supports up to 8 workers (4 Petrol, 4 Diesel) in Phase 1
Account Creation	Super Admin: creates Admin + Worker accounts. Admin: creates Worker accounts only
Account Deletion	Super Admin: can delete Admin + Worker accounts. Admin: cannot delete any account

Status Toggle	Admin can set a worker Active or Inactive without deleting the record
Profile Photo Upload	JPEG or PNG, stored locally or on S3
Shift Assignment	Each worker is assigned Morning, Evening, or Night shift
Fuel Type Assignment	Each worker is assigned Petrol or Diesel

3.5 Daily Entries

Full read access for Super Admin. Full edit for Admin. Workers do not access this module directly — they submit entries via Sales Entry.

Feature	Specification
Entry Fields	Date, Time, Shift, Worker Name, Fuel Type, Tank Stock, Nozzle Reading, Qty Sold, Remaining Stock, Notes
Calendar Navigation	Browse entries by date; date picker for single or range selection
Filter Options	Filter by Date, Worker, Fuel Type, Nozzle
Auto-calculations	Remaining Stock auto-computed from Tank Stock – Qty Sold
Audit Trail	Each entry records who created/modified it and when
Historical Browse	Admin can view any past entry; Super Admin can view but not edit

3.6 Shift Management

Super Admin views all shift records. Admin can create, edit, and delete shifts and assignments. Workers can view their own assigned shift details only.

Feature	Specification
Shift Types	Morning, Evening, Night — all three active in Phase 1
Shift Record Fields	Start Time, End Time, Assigned Workers, Total Fuel Sales, Total Cash Collection
Worker Assignment	Admin assigns workers to a shift; each worker has one active shift at a time
Shift Sales Rollup	Total fuel sold and cash collected auto-aggregated from all entries in the shift
Worker View	Workers see: shift name, start/end time, assigned nozzle, fuel type. Read-only.
Overlap Prevention	System warns if a worker is assigned to two overlapping shifts

3.7 Expense Tracking

Super Admin views all expenses. Admin can create, edit, and delete expense records. Workers have no access.

Feature	Specification
Expense Fields	Expense Name, Category, Amount (INR), Date, Added By, Notes
Categories	Water / Drinking Water, Electricity, Generator Fuel, Miscellaneous / Other
Custom Category	Admin can add custom categories beyond the default four
Daily Total	Dashboard shows running expense total for current day
Monthly Rollup	Expenses rolled up by category for monthly reports
Added By Audit	System records which user created each expense entry

3.8 Generator Management

Super Admin views records. Admin can create, edit, and delete sessions. Workers have no access.

Feature	Specification
Fuel Allocation Logging	Admin records how much fuel was allocated to the generator from tank stock
Consumption Session	Each generator run is a session: start time, end time, fuel consumed
Stock Deduction	Generator fuel allocation reduces the relevant fuel tank stock automatically
Usage Duration	System calculates session duration from start and end timestamps
Difference Calculation	Allocated Fuel – Consumed Fuel = Difference; flagged if non-zero
Historical Log	Full history of generator sessions with timestamps and attribution

3.9 Sales Analysis

Super Admin views all analytics. Admin can view and also update fuel rates. Workers have no access to this module.

Feature	Specification
Sales Metrics	Quantity Sold (litres), Total Revenue (INR), Rate Per Litre, Shift Sales, Daily Sales, Monthly Sales

Rate Management	Admin sets current Petrol and Diesel rate per litre; changes take effect immediately
Rate History	Full timestamped log of every rate change; used to back-calculate historical revenue
Revenue Formula	Total Amount = Quantity Sold × Rate Per Litre at time of sale
Shift Breakdown	Sales aggregated per shift with start/end time and worker count
Daily Breakdown	All shifts rolled into a daily total with comparison to previous day
Monthly View	Calendar-based monthly revenue summary with day-by-day drill-down
Filter & Export	Filter by date range, fuel type; export to CSV or PDF

3.10 Reports

Super Admin can view and export all report types. Admin can generate, view, and export all reports. Workers have no access.

Report Type	Contents	Trigger
Daily Report	Fuel sold (Petrol & Diesel), closing stock, expenses, worker activity, shift summary	Manual or end-of-day auto-generation
Monthly Report	Cumulative sales, total expenses by category, profit estimate, worker productivity	Manual or month-end auto-generation
Worker Report	Individual attendance, shift history, assignments, total sales handled per worker	On-demand by Admin/Super Admin

All reports support export in PDF and CSV formats. Exported reports respect the user's selected language setting.

3.11 Worker-Exclusive Pages

3.11.1 Check In

Feature	Specification
Check-In Action	Worker taps 'Check In' button; system records timestamp and links to active shift
Check-Out Action	Worker taps 'Check Out'; system records timestamp and closes attendance session
Single Active Session	System prevents duplicate check-in if worker is already checked in

Today's Log	Worker sees their own check-in and check-out times for the current day
Geo / Device Tag	Optional: device identifier logged for audit purposes (Phase 2 for geo)

3.11.2 My Attendance

Feature	Specification
Attendance History	List of all past shifts with check-in time, check-out time, and duration
Absence Flag	Shifts where no check-in was recorded are marked as Absent
Late Arrival Flag	Check-ins after shift start time are flagged as Late
Filter by Month	Worker can filter their attendance history by calendar month
Read-Only	Worker can view but not edit attendance records

3.11.3 Sales Entry

Feature	Specification
Entry Scope	Worker can only submit entries for their assigned nozzle and active shift
Entry Fields	Fuel Type (auto-filled), Nozzle (auto-filled), Quantity Sold, Notes
Validation	Quantity must be a positive number; submission blocked if shift not active
Stock Impact	Approved entry triggers automatic stock deduction in Fuel Management
Submission History	Worker can view their own submitted entries for the current shift/day
No Delete	Workers cannot delete their own submitted entries; only Admin can delete

4. Non-Functional Requirements

4.1 Performance

- Dashboard must load within 2 seconds on a standard broadband connection
- All CRUD operations must complete and reflect in the UI within 1 second
- Report generation for a month's data must complete within 5 seconds

4.2 Security

- All passwords stored using bcrypt hashing (cost factor ≥ 12)
- JWT tokens expire after 8 hours; refresh token mechanism optional in Phase 2
- All API endpoints validate the caller's role before processing
- Session auto-logout after 30 minutes of inactivity
- HTTPS enforced for all traffic; HTTP redirects to HTTPS
- Full activity log maintained for audit: user, action, timestamp, affected record

4.3 Usability

- System must be fully usable on desktop and tablet browsers (minimum 768px width)
- Language switcher visible in top navigation bar on all screens
- Error messages displayed in the user's selected language
- All destructive actions (delete) require a confirmation dialog

4.4 Reliability

- Target uptime: 99.5% (excluding scheduled maintenance)
- Database backup support: daily automated backups
- System must handle concurrent sessions for all 8 workers + Admin simultaneously

4.5 Localization

- Phase 1: English and Hindi fully supported
- All UI strings externalized to JSON locale files (en.json, hi.json)
- Date format: DD/MM/YYYY for Indian locales
- Currency format: INR with comma as thousands separator
- Unicode-compatible fonts loaded per language (Noto Sans family)

5. System Operational Flow

The following describes a standard operational day in BPMS:

Step	Action
1	User logs in with username and password; system validates credentials, identifies role, and grants access.
2	Dashboard loads with role-specific widgets: stock gauges, alerts, shift status, and sales summaries.

3	Worker navigates to Check In and records shift start.
4	Worker records opening nozzle reading in Nozzle Management for their assigned nozzle.
5	Worker submits fuel sales entries via Sales Entry throughout the shift.
6	Stock balances update automatically with each validated sales entry.
7	Admin monitors dashboard for low-stock alerts or difference flags and takes corrective action.
8	At shift end, worker submits closing nozzle reading; system auto-calculates fuel sold.
9	Worker records Check Out via Check In page.
10	Admin logs any expenses and generator fuel usage for the shift.
11	System generates shift-level summary; Admin reviews and resolves any difference flags.
12	At day end, Admin or system auto-generates Daily Report; Super Admin reviews read-only.

6. Technology Stack

Layer	Primary Choice	Alternative
Frontend	React.js / Next.js	Plain HTML + CSS + JS
Backend	Node.js + Express	Python Django / Flask
Database	PostgreSQL	MySQL
Authentication	JWT + bcrypt	Session-based auth
i18n Library	react-i18next	LinguiJS
File Storage	Local / AWS S3	Firebase Storage
Deployment	Linux VPS / AWS EC2	Heroku / Render
Fonts	Noto Sans (Google Fonts)	Self-hosted subsets

7. Database Schema Overview

Table	Primary Purpose
Users	Authentication credentials, hashed passwords, roles
Workers	Worker profiles, fuel type, shift, photo, status

FuelStock	Real-time and historical stock levels per fuel type
TankEntries	Refill and deduction records per tank with timestamps
NozzleReadings	Opening and closing readings per shift per nozzle
Sales	All sales transactions: quantity, rate, amount, worker, shift
Expenses	Expense records by category, amount, date, and user
Attendance	Worker check-in/check-out per shift with timestamps
Shifts	Shift definitions and worker-shift assignments
Reports	Cached or generated report metadata and export links
GeneratorLogs	Generator fuel allocation and consumption sessions
RateHistory	Historical log of all Petrol and Diesel rate changes
ActivityLogs	Full audit trail: user, action, record ID, timestamp

8. Future Feature Roadmap

Feature	Target Phase
SMS Alerts — low stock & daily summaries	Phase 2
WhatsApp-based report delivery	Phase 2
Native Android / iOS mobile application	Phase 2
Automated cloud backup & disaster recovery	Phase 2
QR Code-based worker attendance	Phase 3
Biometric attendance integration	Phase 3
AI-based sales prediction & trend analysis	Phase 3
Multi-pump (multi-branch) support	Phase 4
Gujarati, Marathi, Punjabi language support	Phase 2
Bengali, Tamil, Telugu language support	Phase 3

RTL / Urdu language support	Phase 4+
-----------------------------	----------

9. Assumptions & Constraints

9.1 Assumptions

- Internet connectivity is available at the Bharat Petroleum premises at all times.
- Staff will receive basic onboarding training before system go-live.
- Fuel rates are updated manually by the Admin whenever market prices change.
- Photo uploads will be stored in JPEG or PNG format only.
- Maximum 8 concurrent workers in Phase 1; schema is extensible for future growth.

9.2 Constraints

- System must be accessible on desktop and tablet browsers (minimum 768px viewport).
- Phase 1 delivery excludes native mobile app and biometric features.
- System must comply with applicable Indian data protection regulations.
- All financial values are in INR; multi-currency is out of scope.

10. Document Sign-Off

Upon review and approval, authorized signatories confirm acceptance of this PRD as the baseline for BPMS development.

Role	Name	Signature & Date
Product Manager	_____	_____
Technical Lead	_____	_____
Business Owner	_____	_____
Super Admin / Approver	_____	_____